

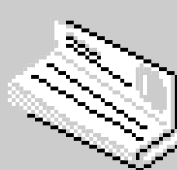
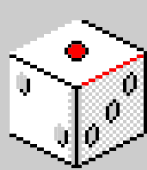
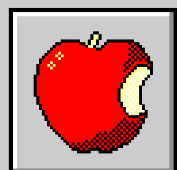
Climate Guardians

Ameera Sofia Binti Zalhan
Nur Iman Binti Mohamad Zahari
Koo Xuan
Ling Yu Qian

B23CS0080
A23CS0158
A23CS0300
A23CS0301



PROGRAMMING TECHNIQUE II



11:11PM

INTRODUCTION

Climate change is a long-term change in weather patterns caused mainly by human activities, such as burning fossil fuels, cutting down forests, and industrial processes. These actions lead to global warming, which results in major changes to weather, rising sea levels, and damaging ecosystems. These changes threaten food production, water availability, health, and people's lives, especially in poorer countries and developing regions. To tackle this issue, the international community created Sustainable Development Goal 13 (SDG 13), called "Climate Action." This goal focuses on urgent efforts to reduce greenhouse gas emissions and protect vulnerable communities from the impacts of climate change. Climate action involves various strategies to adapt to and mitigate the effects of climate change, aiming for a more sustainable and resilient future. Our project will look into the main problems and challenges in climate action, finding gaps that need to be addressed. We will propose innovative and effective solutions across different sectors and levels. Additionally, we plan to create a C++ program to calculate carbon footprints, which will be a useful tool for supporting climate action efforts.



PROBLEM BACKGROUND

Climate change is causing higher temperatures and more frequent extreme weather events, threatening human lives, infrastructure, and ecosystems. Problems like heat waves, melting ice caps raising sea levels, severe storms, floods, and droughts are all linked to rising temperatures. To address these issues, we need sustainable methods to reduce our carbon footprint, lessen negative impacts, and move toward a carbon-free economy.

Fighting climate change is crucial for meeting sustainable development goals, protecting ecosystems, and ensuring fair access to resources and opportunities for everyone. Vulnerable communities are especially harmed, and social inequalities are worsened. Climate change also poses serious health risks, like heat-related illnesses, diseases spread by insects, and food insecurity. By reducing carbon emissions and promoting sustainable practices, we can improve air quality, boost food security, and protect public health. Building resilience to climate impacts involves investing in sturdy infrastructure, early warning systems, and disaster preparedness to minimize damage from extreme weather. Cooperation among nations, businesses, and society is essential to create tools that help everyone monitor their environmental impact and adopt sustainable practices.



PROPOSED SOLUTION

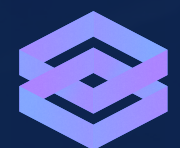
✓ **USER INTERACTION**

✓ **CARBON FOOTPRINT TRACKING**

✓ **FOUNDATIONS FOR GOAL SETTING**

✓ **EDUCATIONAL RESOURCES FRAMEWORK**

✓ **DATA ANALYTICS FRAMEWORK**



PROPOSED SOLUTION

✓ **USER INTERACTION**

✓ **CARBON FOOTPRINT TRACKING**

✓ **FOUNDATIONS FOR GOAL SETTING**

✓ **EDUCATIONAL RESOURCES FRAMEWORK**

✓ **DATA ANALYTICS FRAMEWORK**



KEY STAKEHOLDERS

Addressing climate change involves many groups with different interests. Governments create and enforce policies to reduce emissions and promote renewable energy, affecting businesses, communities, and individuals. NGOs, advocacy groups, and grassroots movements raise awareness and push for policy changes, often representing marginalized communities who are most affected by climate change.

Local communities in vulnerable areas, such as coastal regions and small islands, face direct impacts like rising sea levels and severe weather. Indigenous peoples, with their deep connection to the land, are especially affected. Young people are demanding more action on climate change since they will face the long-term effects of today's decisions. Considering the views of all these groups is essential for effective climate strategies. Working together can lead to balanced solutions that address environmental, social, and economic needs.





OBJECTIVES



INDIVIDUAL

- Helps understand how daily actions impact the environment.
- Encourages reducing carbon footprint and combating climate change.
- Allows setting and tracking personal environmental goals.
- Can save money on energy bills.
- Promotes healthier habits like walking or biking.
- Connects with others who care about the environment.

COMMUNITY

- Raises awareness of carbon emissions within the community.
- Encourages collective responsibility for the environment.
- Empowers community members to work together on sustainability.
- Strengthens community bonds and resilience to climate impacts.
- Supports policies for environmental protection.
- Educates the community on sustainable living practices.

GOVERNMENT

- Provides data for environmental policies and actions.
- Targets high-emission areas for more effective climate action.
- Checks the effectiveness of climate change strategies.
- Involves people in climate action through footprint tracking.
- Helps enforce environmental regulations.
- Facilitates government actions to protect the environment.

EDUCATION

- Teaches students about climate science and sustainability.
- Connects lessons to real-world issues through projects.
- Schools set an example by adopting green practices.
- Uses tools like carbon footprint calculators for education.
- Encourages mindful actions to protect the planet for the future.



SCOPES



MODULARITY

- Breaking down the software into separate modules.
- Each module handles specific tasks (like tracking carbon footprints).
- Modules for tracking, raising awareness, and collaboration.

ENCAPSULATION

- Hiding how each module works internally.
- Protects other modules from changes inside one.
- Only showing necessary ways to use each module.

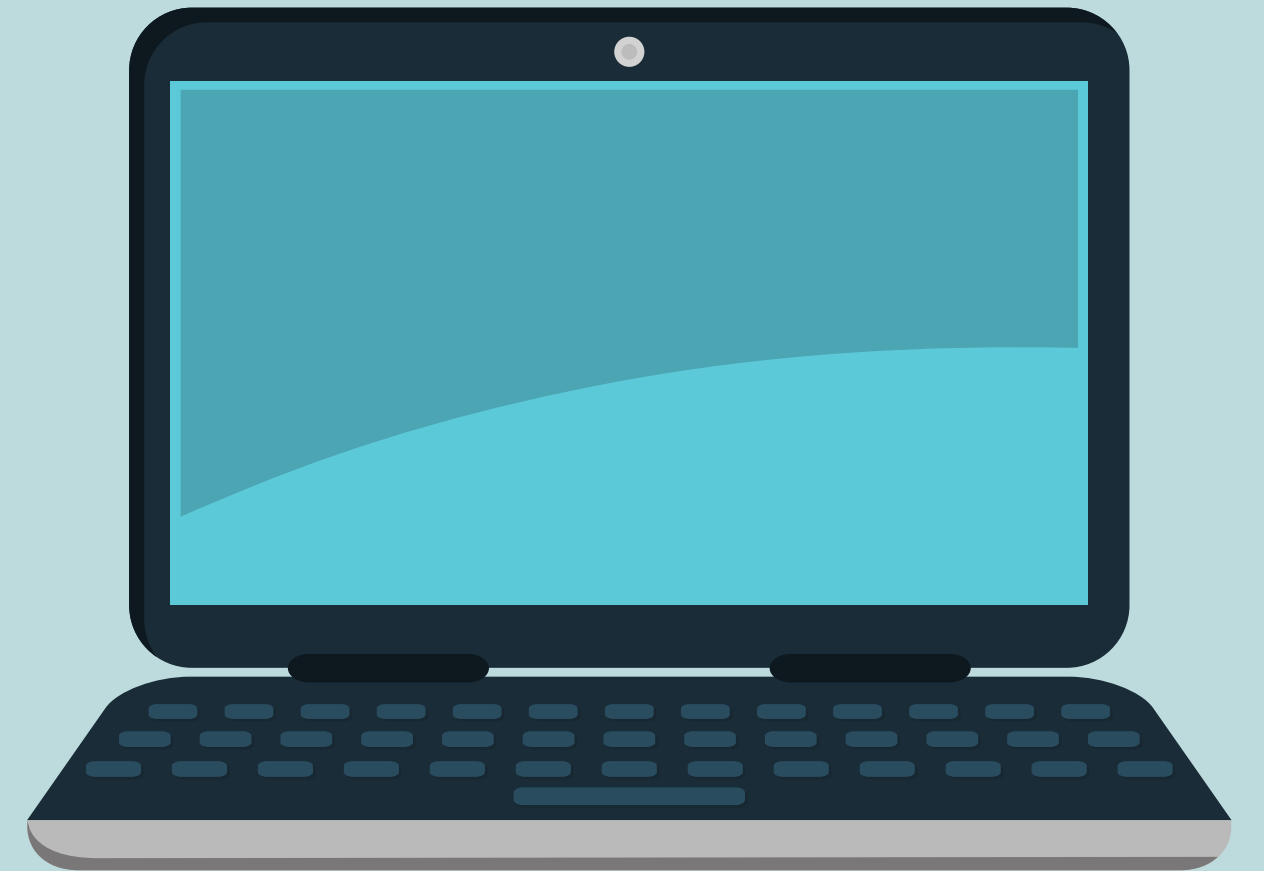
INHERITANCE

- Making new modules that build on existing ones.
- Saves time by reusing what's already made.
- Special modules for different parts (like transportation), using a common base

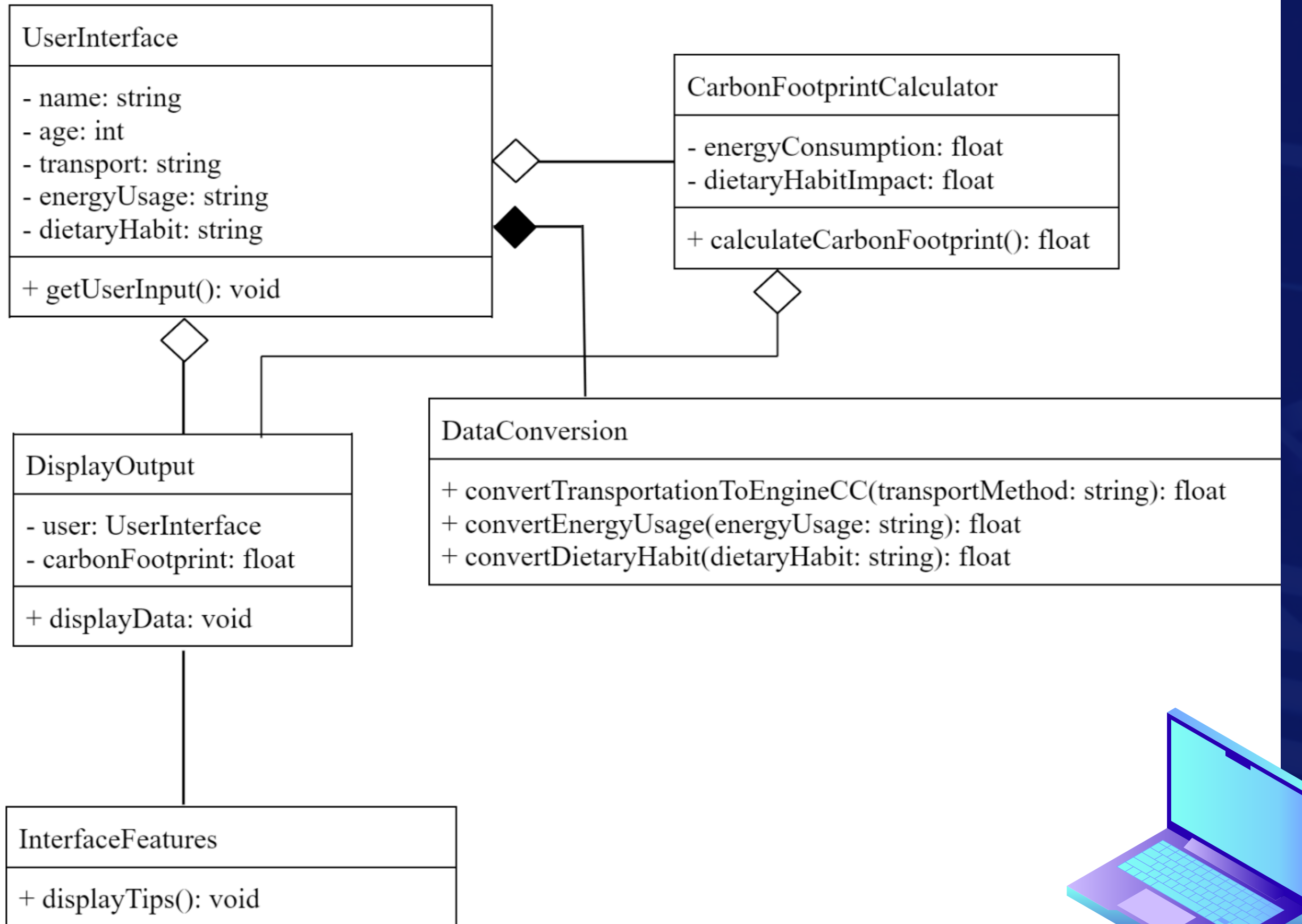
POLYMORPHISM

- Treating different modules in a similar way.
- Helps the software handle various needs easily.
- Using any module interchangeably if they follow the same rules.

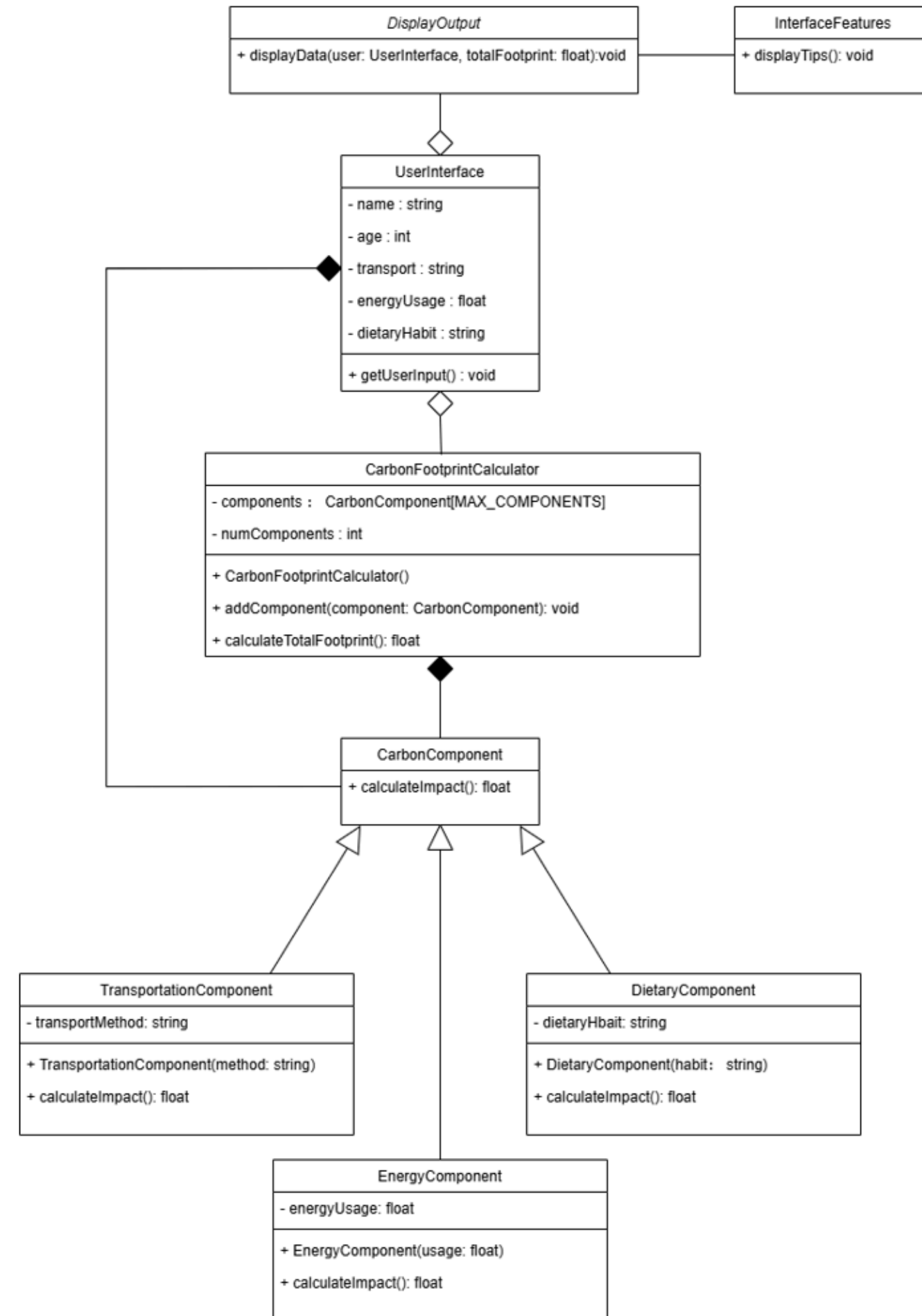
CLASS DIAGRAM



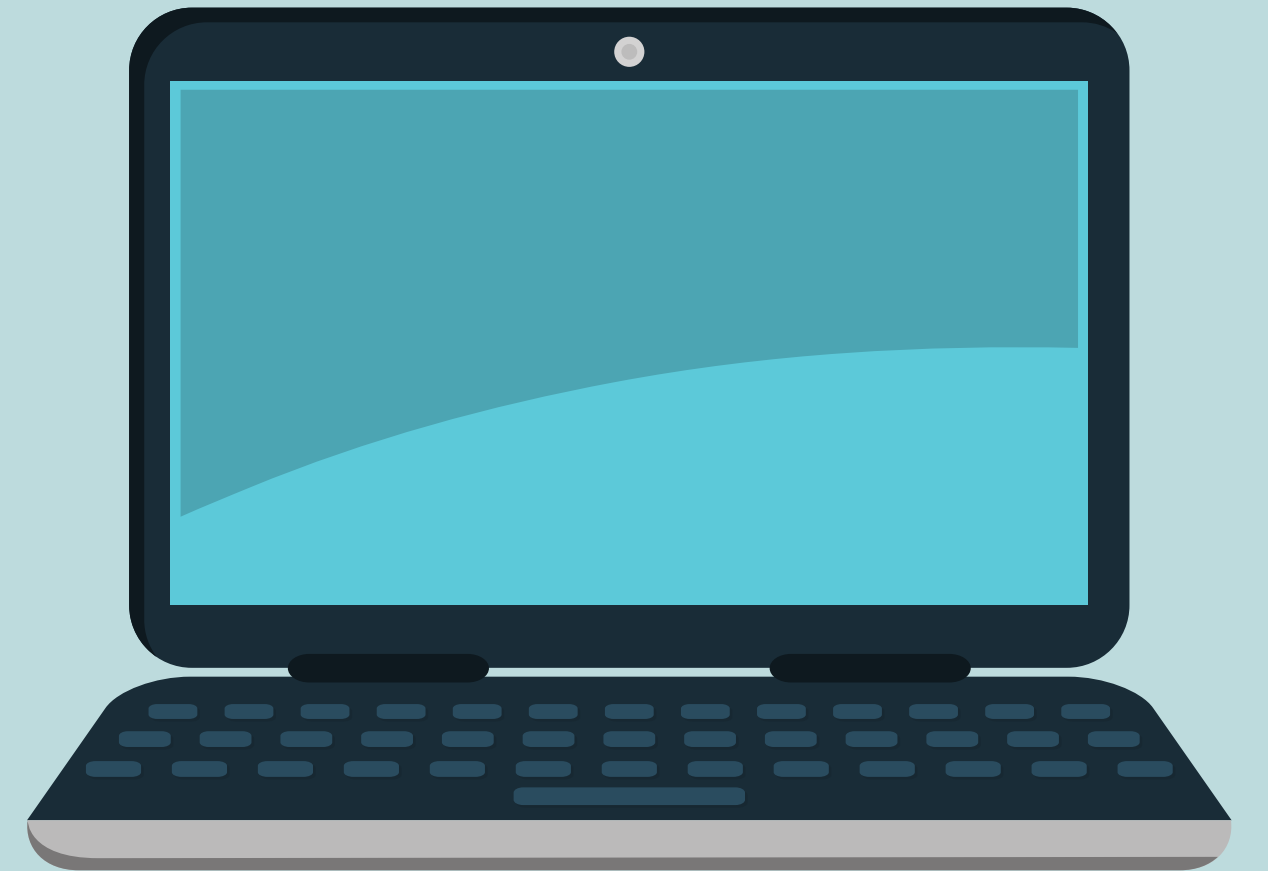
UML Class Diagram



Extended UML Class Diagram (Inheritance and Polymorphism)



CODING



SUMMARY

The proposed system offers numerous benefits by addressing climate change through a comprehensive software solution leveraging Object-Oriented Programming (OOP) principles.

Increased Awareness and Education

- Helps individuals, communities, governments, and educational institutions understand their carbon footprint.

Health and Lifestyle Improvements

- Promotes healthier lifestyles by reducing carbon emissions and encouraging sustainable habits like walking or biking.

Resilience and Adaptation

- Strengthens community bonds and promotes overall sustainability.

This proposed system empowers individuals, communities, governments, and educators to take effective climate action, reduce carbon emissions, and promote sustainable development. By leveraging OOP principles, the system ensures flexibility, adaptability, and ease of maintenance, making it a robust tool in the fight against climate change.





Thank you!